Lesson 03 Demo 04

Creating a Component That Re-Renders in React

Objective: To create a React component that re-renders when a prop changes, using Vite for a faster development experience

Tools required: Visual Studio Code and Node.js

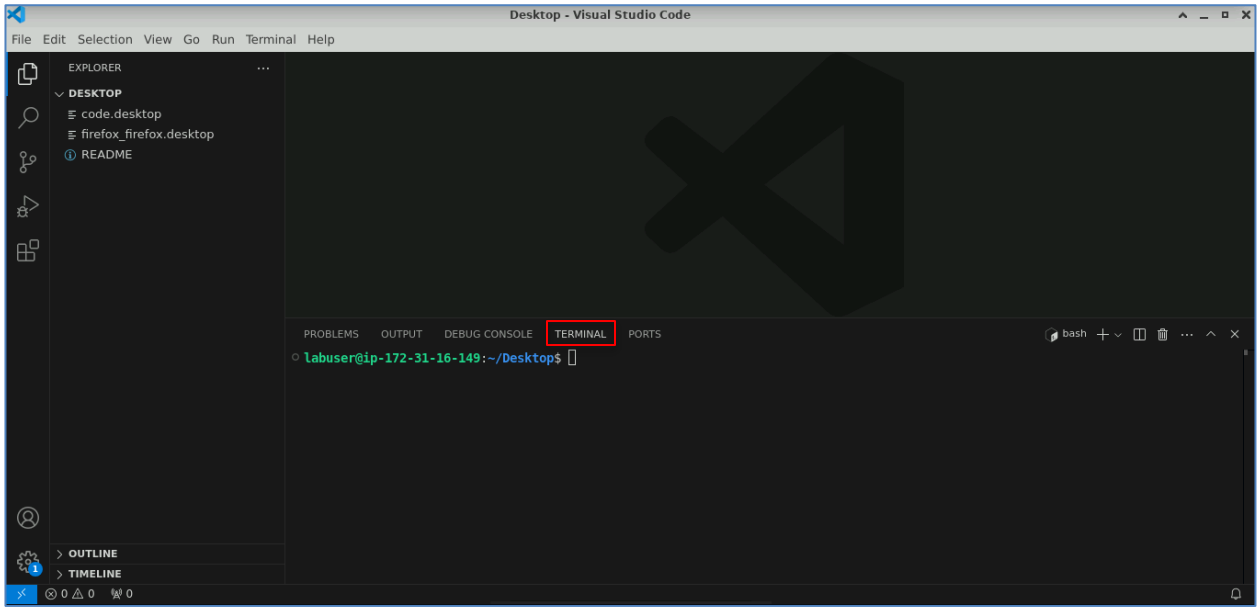
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create a React component
3. Import and use the component
4. Run and verify the application

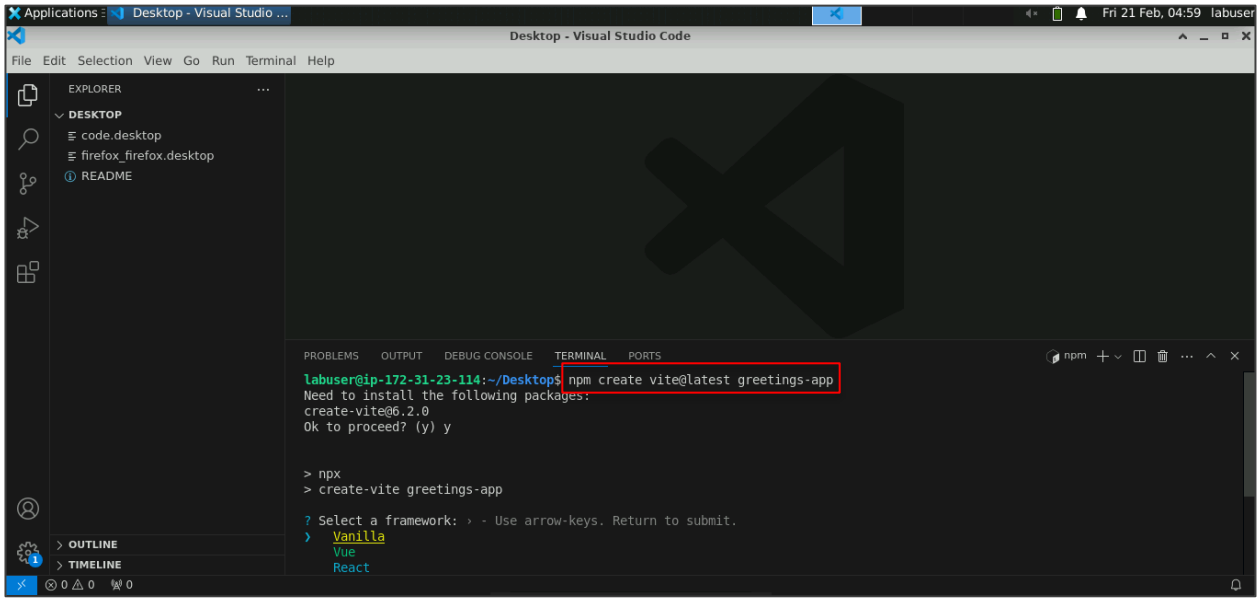
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

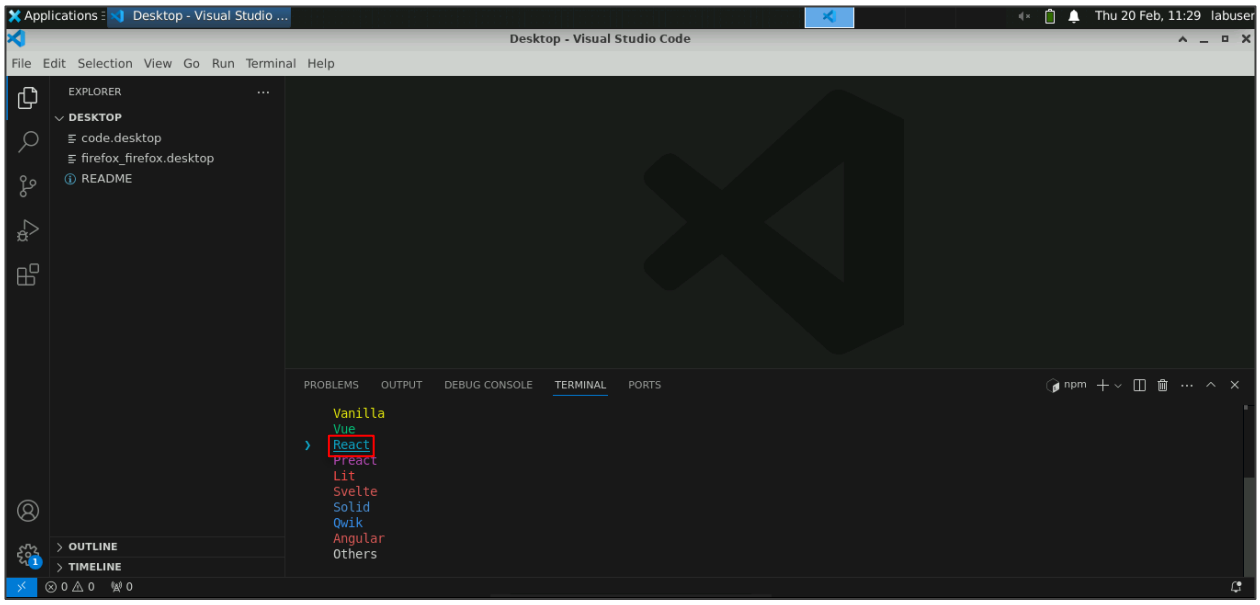


1.2 Run the following command to create a new React project using Vite:

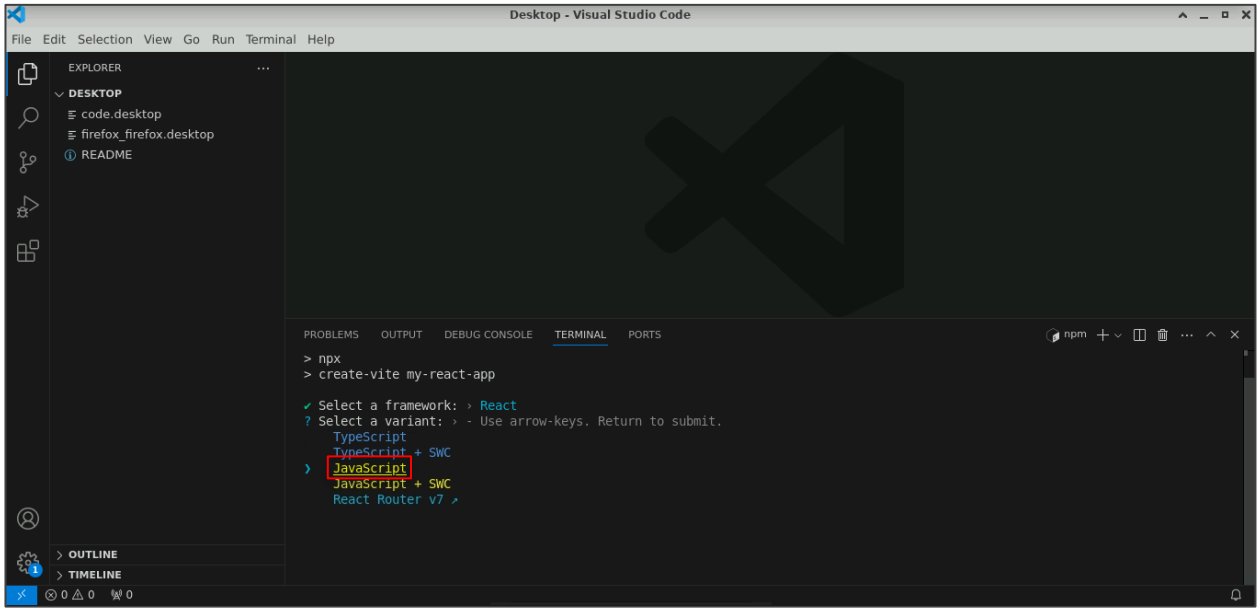
npm create vite@latest greetings-app



1.3 Select **React** and press enter

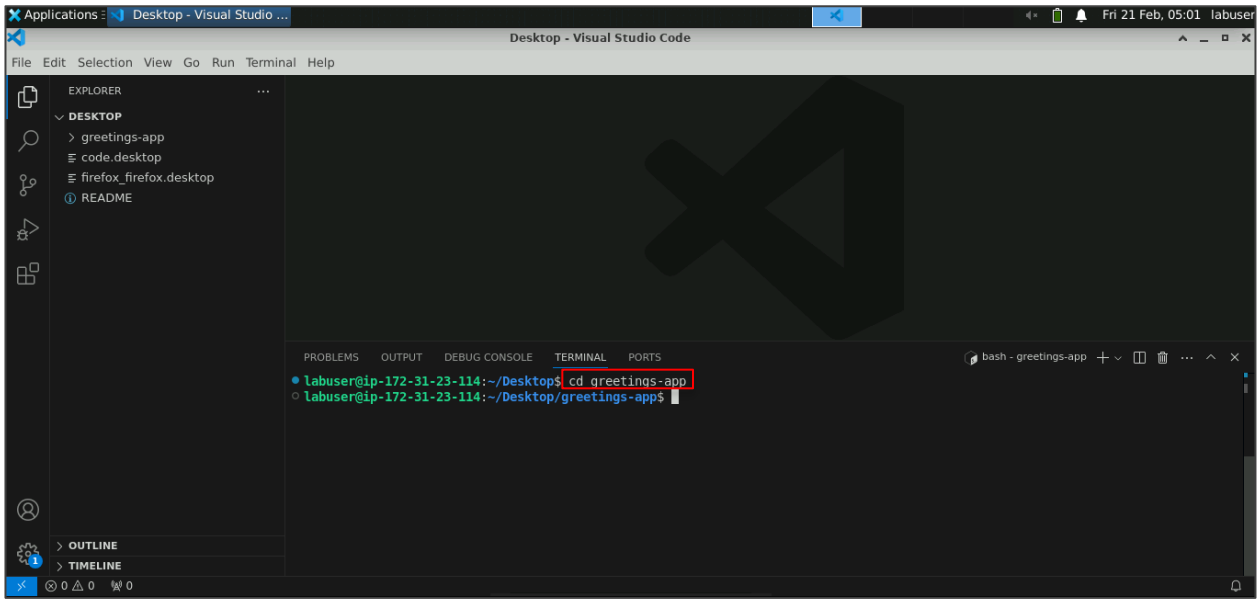


1.4 Select **JavaScript** and press enter



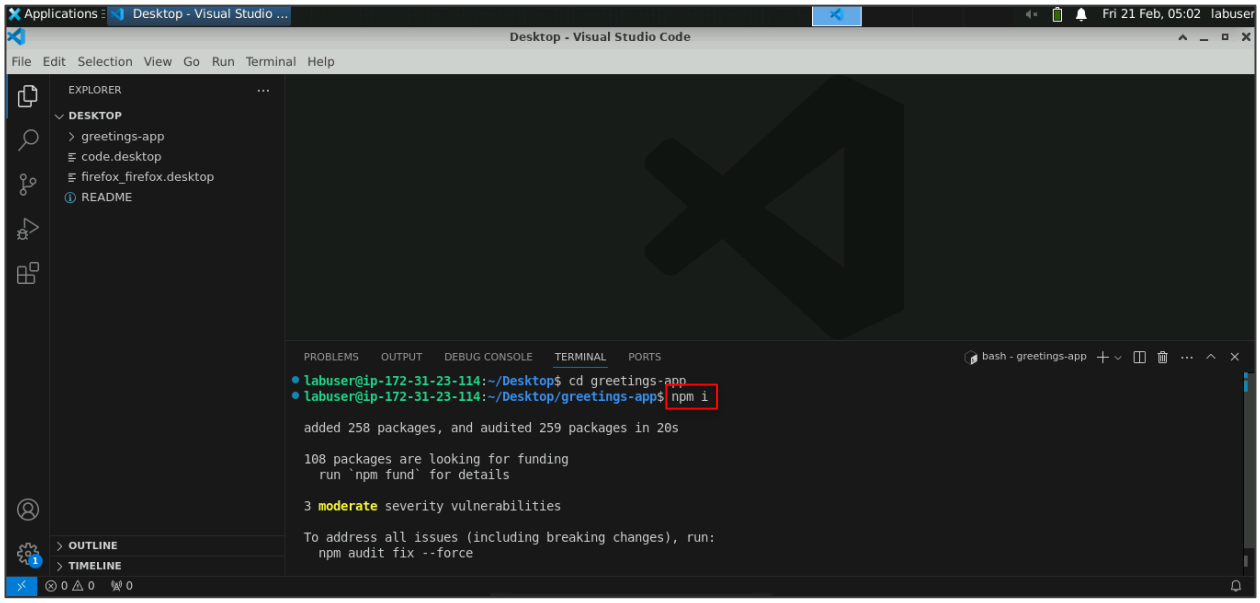
1.5 Run the following command to navigate to the project folder:

cd greetings-app



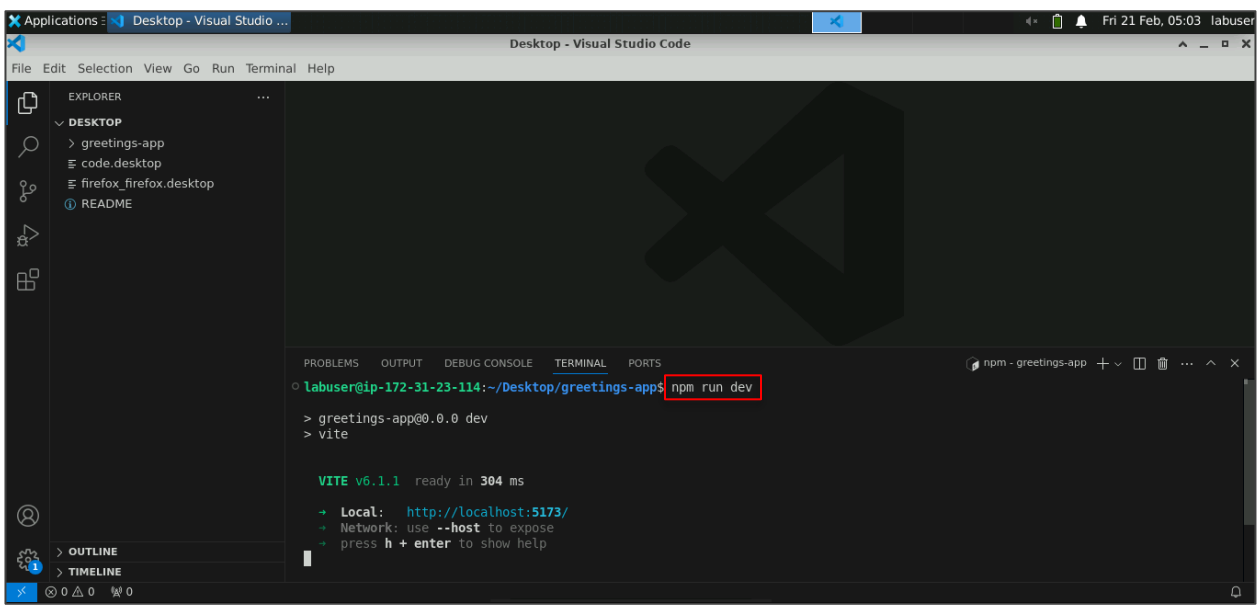
1.6 Run the following command to install dependencies:

npm i



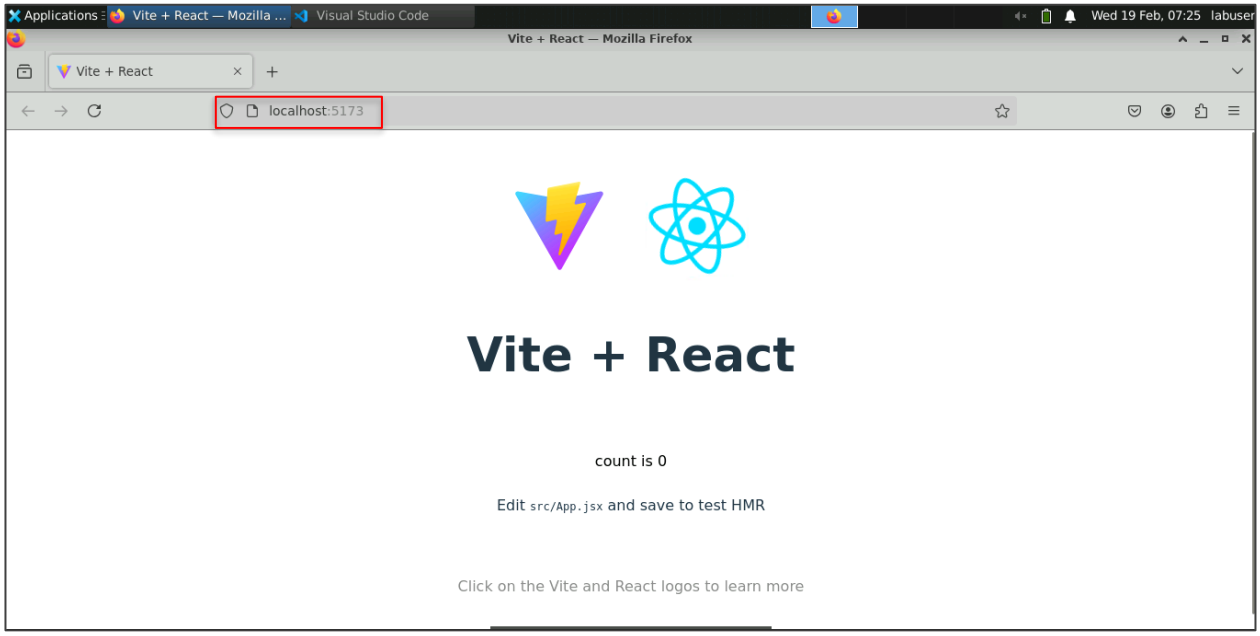
1.7 Run the following command to start the development server:

npm run dev



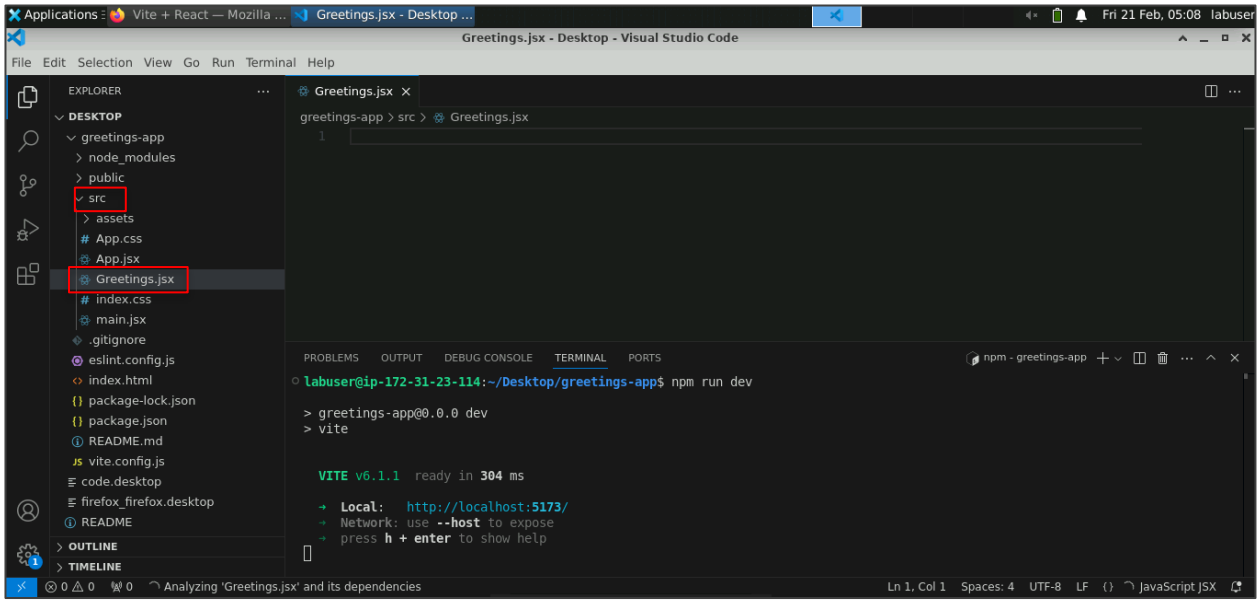
1.8 Navigate to the web browser and paste the link below:

http://localhost:5173



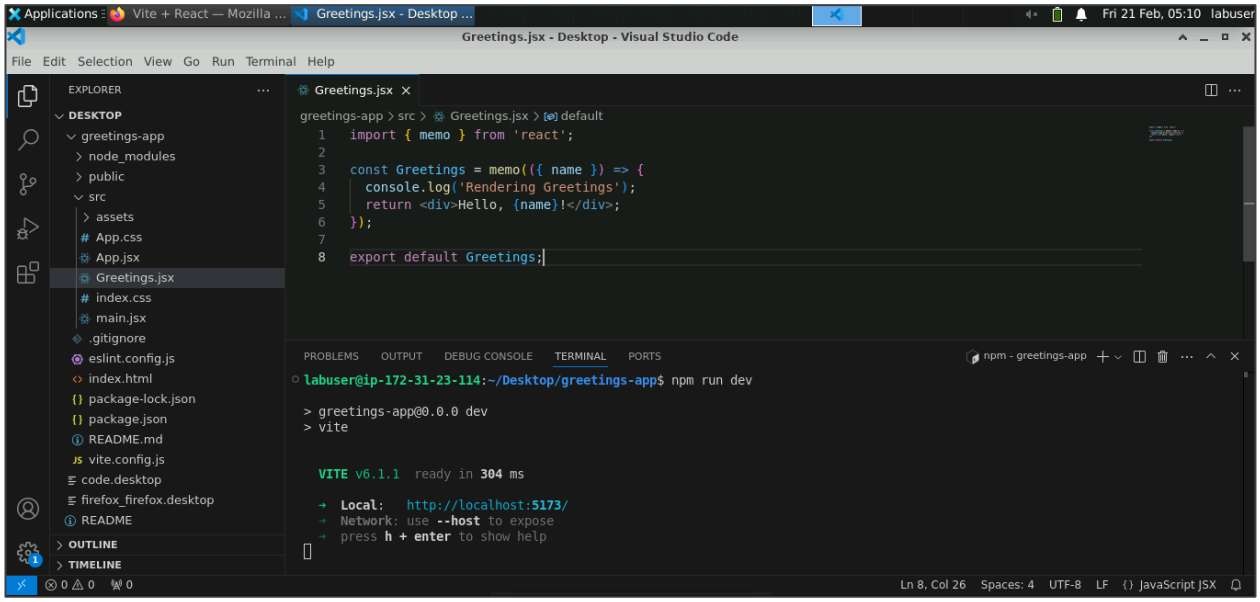
Step 2: Create a React component

2.1 Inside the **src** folder, create a new file named **Greetings.jsx**



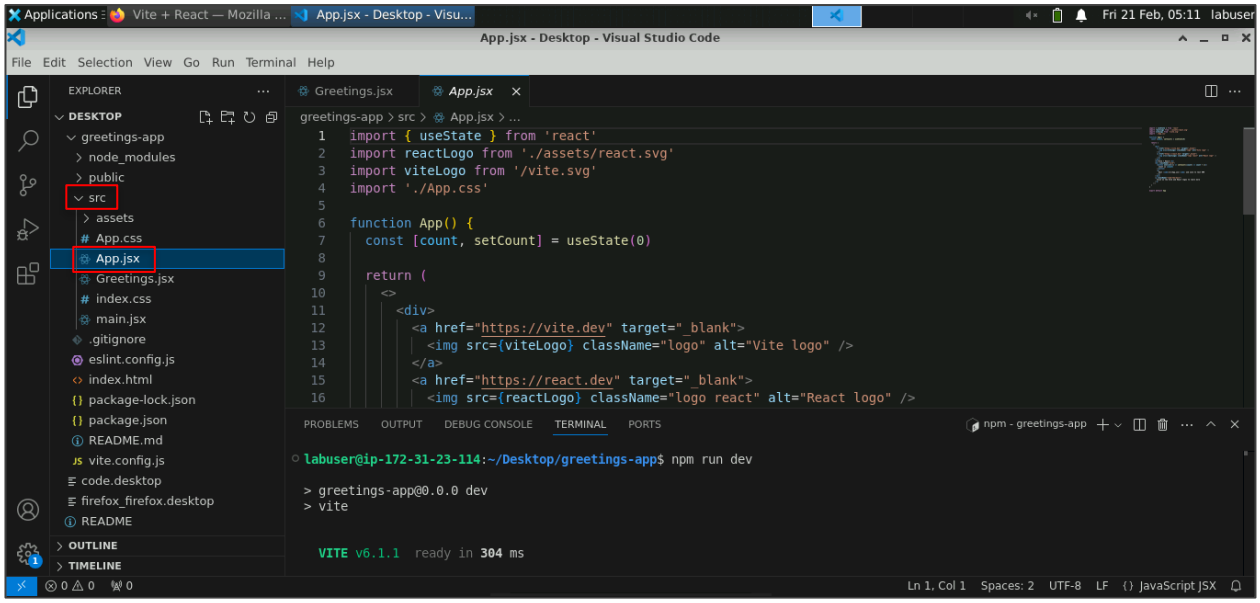
2.2 Add the following code to **Greetings.jsx**

```
import { memo } from 'react';

const Greetings = memo(({ name }) => {
  console.log('Rendering Greetings');
  return <div>Hello, {name}!</div>;
});

export default Greetings;
```



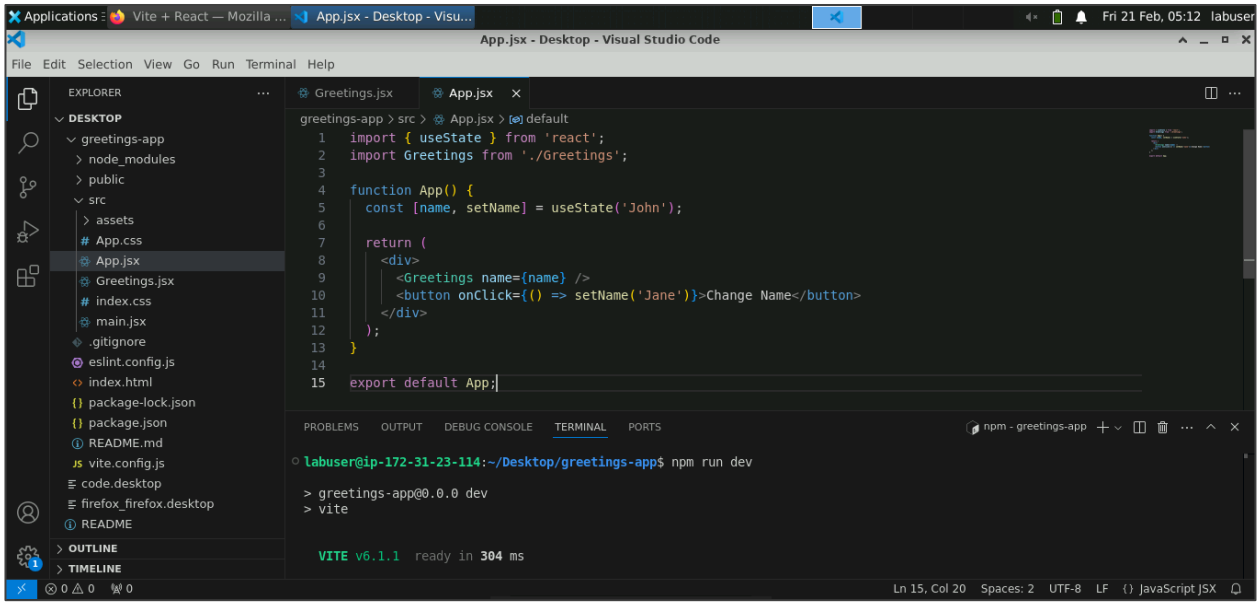
Step 3: Import and use the component

3.1 Open the **App.jsx** file inside the **src** folder



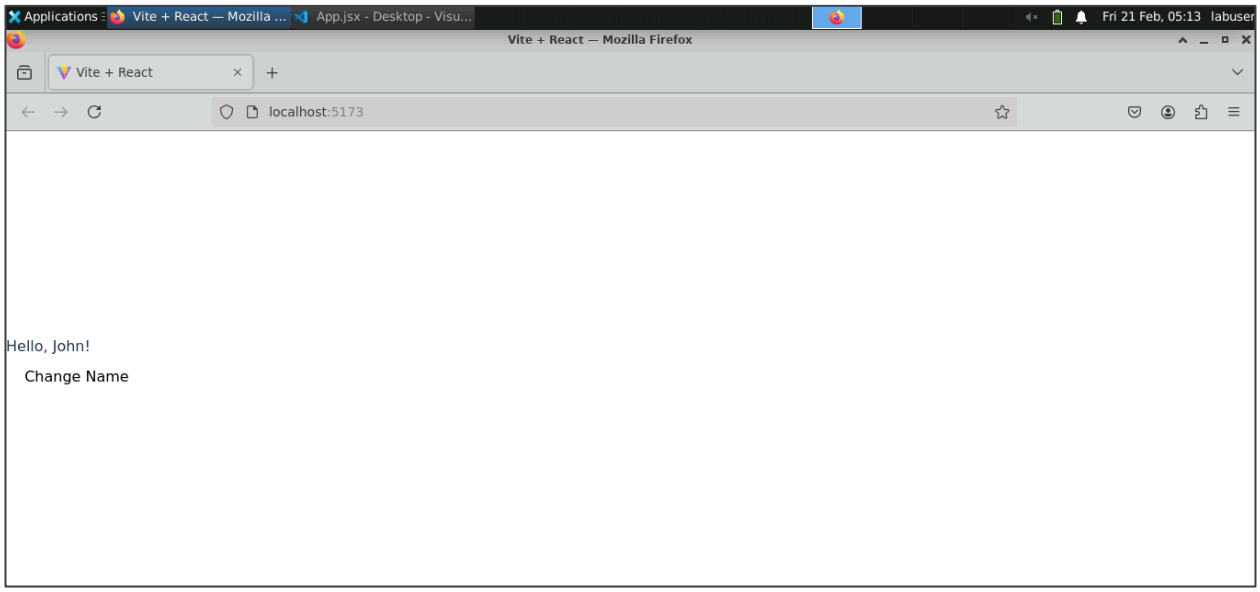
3.2 Paste the code below to include the new component in **App.jsx**:

```
import { useState } from 'react';
import Greetings from './Greetings';

function App() {
  const [name, setName] = useState('John');

  return (
    <div>
      <Greetings name={name} />
      <button onClick={() => setName('Jane')}>Change Name</button>
    </div>
  );
}

export default App;
```

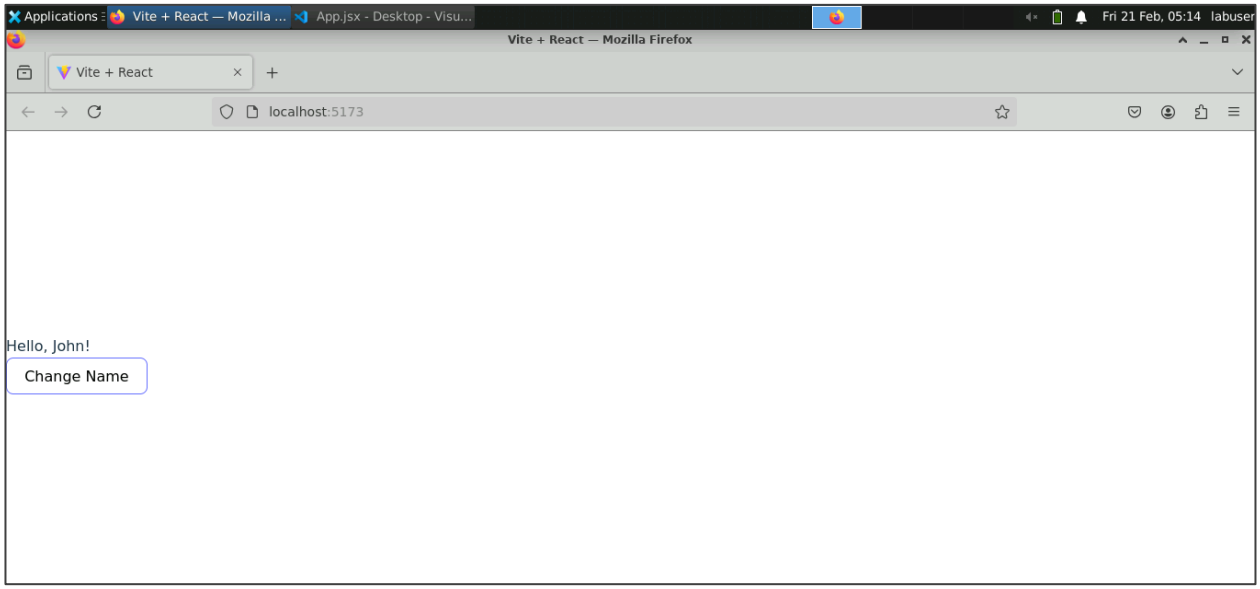


Step 4: Run and verify the application

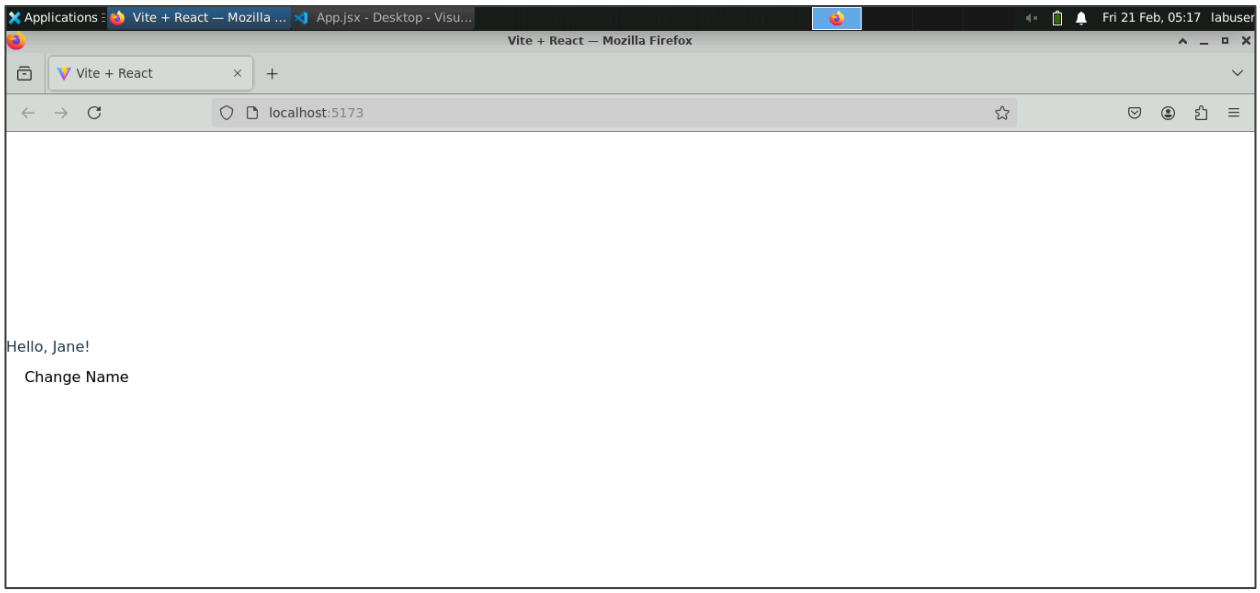
4.1 Navigate to the browser and refresh the webpage



4.2 Click **Change Name** to update the name state to **Jane**



Note: Clicking the **Change Name** button triggers a re-render of the Greetings component with the new value of the name prop.



By following the above steps, you have successfully created a React component that re-renders when a prop changes. Using Vite ensured fast setup and live reloading, while the Greetings component updates dynamically based on state changes in App.jsx.